

Red meat and colon cancer: should we become vegetarians, or can we make meat safer?

The effect of meat consumption on cancer risk is a controversial issue. However, recent meta-analyses show that high consumers of cured meats and red meat are at increased risk of colorectal cancer. This increase is significant but modest (20-30%). Current WCRF-AICR recommendations are to eat no more than 500 g per week of red meat, and to avoid processed meat. Moreover, our studies show that beef meat and cured pork meat promote colon carcinogenesis in rats. The major promoter in meat is heme iron, via N-nitrosation or fat peroxidation. Dietary additives can suppress the toxic effects of heme iron. For instance, promotion of colon carcinogenesis in rats by cooked, nitrite-treated and oxidized high-heme cured meat was suppressed by dietary calcium and by α -tocopherol, and a study in volunteers supported these protective effects in humans. These additives, and others still under study, could provide an acceptable way to prevent colorectal cancer. Copyright © 2011 Elsevier B.V. All rights reserved.